

Monday, 29/8/2022

Lecture 12:

Let G be a group. G is called abelian or commutative if $a * b = b * a \quad \forall a, b \in G$.

Matrix groups: $(M_{n \times m}(\mathbb{Z}), +)$, $(M_{n \times m}(\mathbb{Q}), +)$, $(M_{n \times m}(\mathbb{R}), +)$ are groups.

We have $M_{n \times m}(\mathbb{Z}) \leq M_{n \times m}(\mathbb{Q}) \leq M_{n \times m}(\mathbb{R}) \leq M_{n \times m}(\mathbb{C})$.

Matrix groups are abelian under addition.

Matrix groups under multiplication: Let $GL_n(\mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}), \det(A) \neq 0\}$.

Then, $GL_n(\mathbb{R})$ is a group under matrix multiplication.

We have $GL_n(\mathbb{Q}) \leq GL_n(\mathbb{R})$, and these are non-commutative groups.

We define $SL_n(\mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) \mid \det(A) = 1\}$ clearly. $SL_n(\mathbb{R}) \leq GL_n(\mathbb{R})$.

We have $SL_n(\mathbb{Z}) \leq SL_n(\mathbb{Q}) \leq SL_n(\mathbb{R})$.

Ex: Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in SL_2(\mathbb{Z})$. Then, $O(A)$ is infinite.

$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Then, $O(B) = 4$. $C = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$, $O(C) = 6$.

We have $A = BC$. Thus, B and C both have finite orders, but BC has infinite order. #

Recall that $S^1 = \{z \in \mathbb{C}^* \mid |z|=1\} = \{e^{i\theta} \mid \theta \in \mathbb{R}\}$.

$$M_\infty = \bigcup_{n \geq 1} M_n$$

Thus, every element of M_∞ looks like $e^{2\pi i k/n}$ for some $n \in \mathbb{N}, k \in \mathbb{Z}$.

$$\therefore M_\infty = \{e^{\pi i q} \mid q \in \mathbb{Q}\}.$$

Also, if $x \notin \mathbb{Q}$, then $e^{\pi i x}$ has infinite order in S^1 .

§ Rotations in $\mathbb{R}^2$:

Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = R_\theta \cdot \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$.
Let $\theta \in \mathbb{R}$. Let $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$.

Then, T gives a rotation by an angle θ .

We have $|R_\theta| = 1$. So, $R_\theta \in SL_2(\mathbb{R})$.

If $n \geq 1$, then $R_\theta^n = \begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix} = R_{n\theta}$.

Also, $R_\theta^{-1} = R_{-\theta}$. $R_\theta^{-n} = R_{-n\theta}$ $\forall n \geq 1$

Thus, $R_\theta^k = R_{k\theta} \quad \forall k \in \mathbb{Z}$,

If $\theta = \frac{2\pi}{n}$, then $o(R_\theta) = n$.

Clearly, R_θ has finite order $\Leftrightarrow \theta = \pi \cdot q$, where $q \in \mathbb{Q}$.

We have $\{R_\theta \mid \theta \in \mathbb{R}\}$ is a group under matrix multiplication.

Elements of finite order in $\{R_0 | a \in \mathbb{R}\}$ is the group $\{R_0 | a = \pi \cdot q, q \in \mathbb{Q}\} = \{R_{\pi \cdot q} | q \in \mathbb{Q}\}$.

We define a relation $\sim$ on $\mathbb{R} \times \mathbb{R}$ as follows:

for $x, y \in \mathbb{R}$, $x \sim y$ if $x - y \in \mathbb{Z}$.

If $x \in \mathbb{R}$, then the equivalence class containing x is the set $\{y \in \mathbb{R} \mid y \sim x\} = \{y \in \mathbb{R} \mid y - x \in \mathbb{Z}\}$

$$= \{x+k \mid k \in \mathbb{Z}\} = x + \mathbb{Z}.$$

Let $\mathbb{R}/\mathbb{Z}$ denote the set of all the equivalence classes.

$$\begin{aligned}\text{Then } \mathbb{R}/\mathbb{Z} &= \{x + \mathbb{Z} \mid x \in \mathbb{R}\} \\ &= \{x + \mathbb{Z} \mid x \in [0, 1)\}.\end{aligned}$$

• $\mathbb{R}/\mathbb{Z}$ is a group under the operation:
 $(x_1 + \mathbb{Z}) + (x_2 + \mathbb{Z}) = (x_1 + x_2) + \mathbb{Z}.$

• $\mathbb{Q}/\mathbb{Z} = \{q + \mathbb{Z} \mid q \in \mathbb{Q}\} = \{q + \mathbb{Z} \mid q \in [0, 1) \cap \mathbb{Q}\}$
 is a subgroup of $\mathbb{R}/\mathbb{Z}.$

Every element of $\mathbb{Q}/\mathbb{Z}$ has finite order. $\#$

Thm, we have

$$(1) \quad \mathcal{M}_\infty \leq S^{-1}$$

$$(2) \quad \{R_{\pi \cdot q} \mid q \in \mathbb{Q}\} \leq \{R_0 \mid 0 \in \mathbb{R}\}$$

$$(3) \quad \mathbb{Q}/\mathbb{Z} \leq \mathbb{R}/\mathbb{Z}.$$

#